

# Build Isaac ROS environment

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## Introduction to the Isaac ROS Environment

The Isaac ROS suite, developed and released by NVIDIA, is designed to leverage NVIDIA acceleration on NVIDIA Jetson and standalone GPUs for developing standard robotics applications.

Isaac ROS uses standard ROS interfaces for input and output, making it easy to get started and a drop-in replacement for common CPU ROS implementations familiar to robotics developers.

## System Requirements

| Platform | Hardware                                                                  | Software            | Notes                                                                                                                                                                                        |
|----------|---------------------------------------------------------------------------|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Jetson   | Jetson Orin                                                               | JetPack 6.1 and 6.2 | For best performance, ensure that power settings are configured appropriately. Jetson Orin Nano 4GB may not have enough memory to run many of the Isaac ROS packages and is not recommended. |
| x86_64   | Ampere or higher<br>NVIDIA GPU<br>Architecture with 8 GB<br>RAM or higher | Ubuntu<br>22.04+    | CUDA 12.6+                                                                                                                                                                                   |

## ROS Support

All Isaac ROS packages are designed and tested for compatibility with ROS 2 Humble.

If you are using ROS 1 Noetic, you can use the Isaac ROS NITROS Bridge to integrate Isaac ROS packages for faster performance.

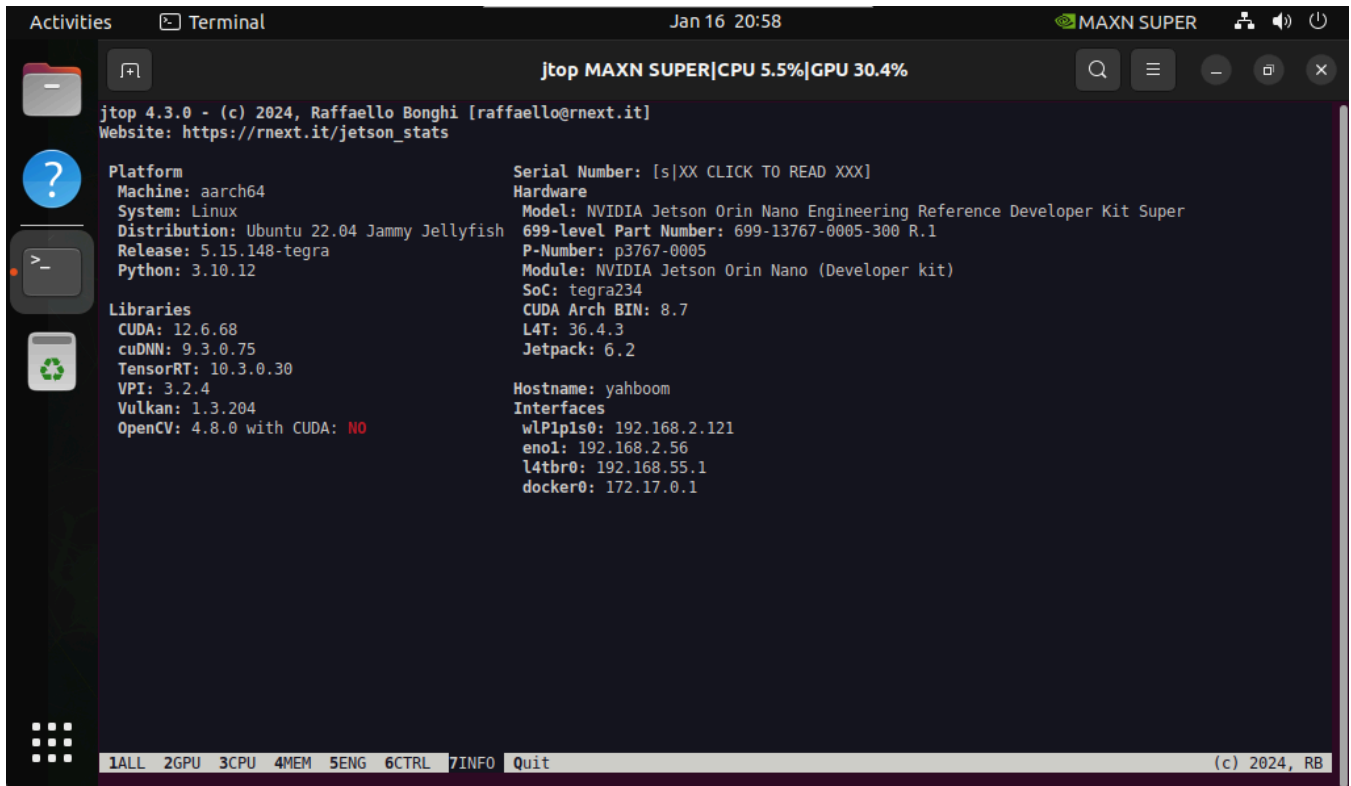
ROS 1 Noetic is not supported on the same system as ROS 2 Humble.

Isaac ROS packages have only been tested with ROS 2 Humble. Other ROS 2 versions are not yet supported.

## Quick Installation

**Note: Installation failure is normal. This environment requires a proxy server to install properly. You can search online for proxy server instructions.**

1. Confirm that your system has JertPack 6.2 installed and set the system power to MAXN SUPER mode.



```
Activities Terminal Jan 16 20:58 MAXN SUPER
jtop MAXN SUPER|CPU 5.5%|GPU 30.4%
jtop 4.3.0 - (c) 2024, Raffaello Bonghi [raffaello@rnext.it]
Website: https://rnext.it/jetson_stats

Platform
Machine: aarch64
System: Linux
Distribution: Ubuntu 22.04 Jammy Jellyfish
Release: 5.15.148-tegra
Python: 3.10.12

Serial Number: [s]XX CLICK TO READ XXX]
Hardware
Model: NVIDIA Jetson Orin Nano Engineering Reference Developer Kit Super
699-Level Part Number: 699-13767-0005-300 R.1
P-Number: p3767-0005
Module: NVIDIA Jetson Orin Nano (Developer kit)
SoC: tegra234
CUDA Arch BIN: 8.7
L4T: 36.4.3
Jetpack: 6.2

Libraries
CUDA: 12.6.68
cuDNN: 9.3.0.75
TensorRT: 10.3.0.30
VPI: 3.2.4
Vulkan: 1.3.204
OpenCV: 4.8.0 with CUDA: NO

Hostname: yahboom
Interfaces
wlp1s0: 192.168.2.121
enol: 192.168.2.56
l4tbr0: 192.168.55.1
docker0: 172.17.0.1

1ALL 2GPU 3CPU 4MEM 5ENG 6CTRL 7INFO Quit (c) 2024, RB
```

## 2. Install Docker

```
# Add Docker's official GPG key:
sudo apt-get update
sudo apt-get install ca-certificates curl gnupg
sudo install -m 0755 -d /etc/apt/keyrings
curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo gpg --dearmor -o
/etc/apt/keyrings/docker.gpg
sudo chmod a+r /etc/apt/keyrings/docker.gpg

# Add the repository to Apt sources:
echo \
"deb [arch="$(dpkg --print-architecture)" signed-by=/etc/apt/keyrings/docker.gpg]
https://download.docker.com/linux/ubuntu \
"$(. /etc/os-release && echo "$VERSION_CODENAME")" stable" | \
sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
sudo apt-get update

sudo apt install docker-buildx-plugin
```

## 3. Add the docker user group

```
sudo usermod -aG docker $USER
newgrp docker
```

## 4. VPI settings

Make sure the NVIDIA Container Toolkit is installed on your Jetson device. Generate the CDI specification using the following command:

```
sudo nvidia-ctk cdi generate --mode=csv --output=/etc/cdi/nvidia.yaml
```

Install the pva-allow-2 package:

```
# Add Jetson public APT repository
sudo apt-get update
sudo apt-get install software-properties-common
sudo apt-key adv --fetch-key https://repo.download.nvidia.com/jetson/jetson-ota-public.asc
sudo add-apt-repository 'deb https://repo.download.nvidia.com/jetson/common r36.4 main'
sudo apt-get update
sudo apt-get install -y pva-allow-2
```

#### 5. Set up the development environment

Clone isaac\_ros\_common under \${ISAAC\_ROS\_WS}/src.

```
cd ${ISAAC_ROS_WS}/src && \
  git clone -b release-3.2 https://github.com/NVIDIA-ISAAC-ROS/isaac_ros_common.git
isaac_ros_common
```

#### 6. Start the Docker container using the run\_dev.sh script:

```
cd ${ISAAC_ROS_WS}/src/isaac_ros_common && \
  ./scripts/run_dev.sh
```

Wait for docker to pull successfully and the environment will be set up.